Studentships in Atmospheric and Planetary Circulations

University of Exeter
College of Engineering, Mathematics and Physical Sciences
Prof. Geoffrey K. Vallis

1. Ph.D. Studentship in Modelling of Planetary Atmospheres

Primary Supervisor: Professor Geoffrey Vallis, Dept. Mathematics.

We invite applications for a Ph.D. studentship in the area of modelling and theory of the general circulation of the atmosphere of planetary atmospheres. The aim of the research will be to understand the possible ranges of such atmospheres and their relation to that of Earth. The work may involve theory and modelling the atmospheres of other solar system planets, such as Venus or Jupiter, and/or planets outside the solar system. The research will investigate how the circulation might depend on such parameters as atmospheric composition, size and rotation rate of the planet and the composition of the atmosphere. You will join a vibrant group of scientists in the Mathematics and Physics department at Exeter engaged in similar geophysical and astrophysical problems.

Application criteria: Applicants should have or expect to achieve at least a 2:1 Honours degree, or equivalent, in mathematics, physics, engineering or related subject. The successful candidate will be a smart, motivated, curious and hard-working independent thinker.

First contact for informal enquiries: Professor Geoffrey Vallis, g.vallis@exeter.ac.uk

How to apply: To apply, you must complete the online web form. You will be asked to submit some personal details and upload a full CV, transcript, covering letter and details of two academic referees. Your covering letter should outline your academic interests, prior research experience and reasons for wishing to undertake this project. You may also upload a thesis or a write-up of a final year project (upload as "research proposal"). For general enquiries please contact Fiona Ayre at emps-pgr-ad@exeter.ac.uk. International applicants are welcome to apply.

2. Ph.D. Studentship in the Circulation and Climate Change

Primary Supervisor: Professor Geoffrey Vallis, Dept. Mathematics.

We invite applications for a Ph.D. studentship in the area of modelling and theory of the response of the circulation of the atmosphere or ocean to climate change, such has occurred in the past (ice ages, hot-house climates) and will occur in the future (global warming). Changes in circulation are poorly understood because they involve feedbacks between different components of the climate system. Thus, they are a considerable scientific challenges as well as being of great societal importance. The research will investigate how such large-scale features, such as the position of the jet stream or the wind-driven circulation of the ocean will change as the planet warms. You will join a vibrant group of scientists in the Mathematics and Physics department at Exeter engaged in similar problems in climate and geophysical fluid dynamics.

Application criteria: Applicants should have or expect to achieve at least a 2:1 Honours degree, or equivalent, in mathematics, physics, engineering or related subject. The successful candidate will be a smart, motivated, curious and hard-working independent thinker.

First contact for informal enquiries: Professor Geoffrey Vallis, g.vallis@exeter.ac.uk

How to apply: For this position please apply through the NERC DTP GW4 web portal. You may wish to contact Prof. Vallis first.